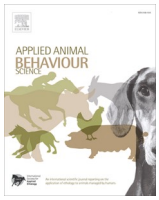
Applied Animal Behaviour Science 305 (2026) 107209



Contents lists available at ScienceDirect

Applied Animal Behaviour Science

journal homepage: www.elsevier.com/locate/applanim



The petting diaries: Investigating owner-cat petting styles, responses and preferences

Kristina O'Hanley [a], David L. Pearl [b], Mary Klinck [c], Lee Niel [b,*]

[a] Department of Animal and Food Science, Texas Tech University, 1308 Indiana Ave, Lubbock, TX, 79409, United States
[b] Department of Population Medicine, Ontario Veterinary College, University of Guelph, 50 Stone Road E., Guelph, ON, N1G 2W1, Canada
[c] Dr. Mary Klinck, DVM, Veterinary Behaviour Consultant, Sainte-Anne-de-Bellevue, QC, Canada

ARTICLE INFO

Keywords:
Feline
Petting
Human-animal interactions
Domestic Cat
Aggression

ABSTRACT

Petting is a common owner-cat interaction typically viewed as enjoyable for most cats. However, little is known about the factors influencing individual cats' responses to petting. This study aimed to identify how owners pet their cats, examine cat responses to petting, and determine risk factors for negative reactions. Cat owners completed a one-day petting log including three petting sessions to record contextual factors (e.g., duration, frequency, cat response, and potential influences). Owner- and cat-related demographics, household details, and lifetime petting responses were collected through an online survey. A total of 66 cat owners completed the petting log. Logistic and mixed logistic regression models were used to evaluate associations between survey data and cat responses. Most cats reportedly enjoyed all three petting sessions (82.7–90.6%), though some displayed warning behaviours (10.9–12.7%). Notably, 32.2% of owners reported being scratched or bitten during petting, and 23.7% reported their cat had scratched or bitten another person during petting. No significant associations were found between evaluated factors and cat responses during the recorded petting sessions. However, lifetime data revealed several significant relationships. The odds of an owner being scratched or bitten during petting were higher when they reported using verbal corrections (e.g., "No!") (OR = 14.2; 95% CI: 2.2–93.7; p = 0.007) or stopping petting (OR = 6.9; 95% CI: 1.4–34.8; p = 0.023) in response to previous incidents. Conversely, providing explorative enrichment in the home (e.g., boxes, tunnels) was protective (OR = 0.15; 95% CI: 0.03–0.81; p = 0.016). The odds of a cat scratching or biting another individual during petting were also increased when owners reported stopping petting after being scratched or bitten previously (OR = 8.0; 95% CI: 2.1–31.2; p = 0.003). These findings suggest that certain owner responses to petting-related aggression may be associated with increased risk of scratching or biting, though causation remains unclear. Owners may either reinforce or inadvertently provoke negative responses, or they may naturally react this way to cats already prone to aggression. Further research is needed to determine directionality. Education on feline body language, early warning signs, and appropriate handling during petting could improve owner understanding of cat preferences and reduce conflict during interactions. Enhancing enrichment opportunities and promoting positive handling techniques may ultimately strengthen the owner-cat relationship and improve feline welfare.

1. Introduction

Petting is a common owner-cat interaction and has the potential to improve cat welfare and build a positive human-animal relationship when performed as preferred by the cat. It has been suggested that owner-cat petting shows some parallels with cat-cat allorubbing (Bradshaw et al., 2012), which is an affiliative social behaviour that tends to be performed between two cats in a social cat group where they rub their cheeks, foreheads, sides, or tails together. It can also be seen directed towards humans, where rubbing against the person is presented as an affiliative behavior, often seen as willingness to engage in petting (Haywood et al., 2021; Soennichsen and Chamove, 2002). In a study completed on cats in an animal shelter, it was found that cats that received gentle petting in combination with verbal communication were less likely to perform behaviours indicating frustration or anxiety, and had reduced occurrences of upper respiratory infections (Gourkow et al., 2014), which indicates that human-cat petting does, in fact, have benefits for cats. Although these results suggest that petting is parallel to

* Corresponding author.
E-mail address: niell@uoguelph.ca (L. Niel).

https://doi.org/10.1016/j.applanim.2026.107209
Received 6 August 2026; Received in revised form 25 September 2026; Accepted 30 September 2026
Available online 5 October 2026
0168-1591/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

allorubbing, with petting being positive for cats, some cats appear to dislike petting, either avoiding the interaction or responding aggressively.

Owner-cat petting interactions can sometimes result in cats showing negative responses, including threats, bites and scratches directed towards the owner, and this is commonly referred to as petting-related aggression. It has been suggested that 15% of all cat bites occur during petting, and these bites can cause serious infections and impair the owner-cat bond (Dire, 1991; Palacio et al., 2007). Of the various forms of human-directed aggression in cats, petting-related aggression has been shown to be the most commonly reported, with one study finding that 21.5% of owners reported petting-related aggression from their cat (Ramos and Mills, 2009). This study also found an association between the cat disliking the petting session and responding in an aggressive manner. The authors suggest that the aggressive response may be due to the physical act of petting rather than negative feelings towards the person performing the petting. This suggests that some cats dislike specific factors associated with petting by humans, and more research is needed to fully understand these factors.

In the limited studies that have examined cat preferences and responses to petting, the findings suggest that cats show fewer negative responses to being petted in the temporal region, between the eyes and ears and show more negative responses to being petted near the tail (Soennichsen and Chamove, 2002). Ellis et al. (2015) also found that cats had more negative responses during petting near the base of the tail when it was performed by an unfamiliar person. In this study, handlers (familiar and unfamiliar) were instructed to stroke areas of the cat's body in eight discrete areas, regardless of previous history or known responses to petting on the cat's body. Additionally, in a preliminary study, cat owners identified areas where their cat enjoyed being petted (based on the cat remaining still or moving their head/body to obtain more petting in a specific area) (Bernstein, 2000, 2007), and reported that cats enjoyed being petted on the top of the head and along the cheeks. Owners in this study also reported cat preferences for particular petting sequences (e.g., pet from head to tail) and for specific petting locations in the house. However, Ellis et al. (2015) reported that order of petting regions did not have an impact on the cat's responses to petting and that more negative responses were seen from owner petting as opposed to petting from an unfamiliar person. It has also been suggested that human-cat interactions may be context and time-dependent, with more allorubbing seen presented towards an owner upon return after a brief departure (Behnke et al., 2021), and thus likely more petting performed during this time.

How an owner responds to their cats' unwanted behaviour might also influence how the cat responds to petting in future events. Studies have reported a significant increase in the odds of aggressive behaviours towards the owner when owners reported using aversive corrections (e.g., yelling at the cat) or negative reinforcement through removal of wanted items or interactions (O'Hanley et al., 2021); however, the direction of this relationship is unknown. Additionally, it has been shown that some cats use social-negative reinforcement, removing themselves from a situation they do not enjoy such as petting (Fritz et al., 2022). Therefore, how both the owner and cat respond during petting or aversive behaviours can set the basis for future owner-cat interactions.

The objectives of the current study were the following: 1) identify how owners pet their cats under normal, routine conditions; and 2) determine how cats respond to different types of petting by the owner. Additionally, we examined whether other cat and owner demographic factors were associated with cat responses to routine petting by their owner. A one-day petting log was used by owners to record and report on their routine petting interactions and their cats' responses to petting throughout the day. This information, along with owner demographics, cat demographics, household information, and historical reports of responses to petting were recorded using an online survey. Based on the results of previous studies on cat petting preferences, we predicted that that owners would report more positive responses for petting near the head and more negative responses for petting near the base of the tail. We also hypothesized that owner-reported use of correction in response to threats and aggression would negatively impact the cat's responses to petting (i.e., negative responses would increase after aversive corrections were used in response to previous negative responses to petting). Information obtained surrounding the nature of the petting interactions (e.g., time of day, length of time for petting session, sequence of petting) will provide exploratory data that can help guide future research regarding owner-cat interactions setting the baseline for what is "normal" in these interactions.

2. Methods

The study protocol was reviewed and approved by the University of Guelph Research Ethics Board (REB# 20–05–007).

2.1. Petting log

The petting log was developed using information gathered from previously published literature on cat petting preferences and behavioural responses to petting (Bradshaw et al., 2012; Ellis et al., 2015; Soennichsen and Chamove, 2002). It included instructions, information on cat petting body regions and responses, and space for owners to record additional information that was relevant during petting sessions. Owners were instructed to record information for three petting sessions: the first petting session would occur during the first 6 h after waking up, the second during the following 6-hour time frame, and the third during the 6-hour time frame after that (e.g., 6 am to 12 pm, 12 pm to 6 pm, 6 pm to 12 am). For each petting session, owners were asked to record the specific time at which it started and ended, the regions they petted on their cat, the responses the cat had to petting, and anything external that happened that might have impacted their cat's response to the petting session. The petting log was piloted by five current cat owners to ensure comprehension and ease of completion prior to distribution. The petting log was attached at the beginning of the promoted survey (See Section Petting Preferences Survey) for owners to download and complete. The full petting log that was available to owners can be found in the Supplementary Materials.

2.2. Petting preferences survey

The surveys used in this study were generated through Qualtrics® Survey Software. To be eligible to participate, individuals had to be at least 18 years of age, residents of Canada or the United States and an owner of a cat that was at least one year old at the time of the survey. A cat owner was defined as someone having primary care and financial responsibility for the cat that was involved in the study. Cat owners were recruited to participate in the survey through online advertisements through various social media platforms (e.g., Facebook and Twitter) and University of Guelph websites. The survey was launched July 13th, 2020 and was available until October 4th, 2020.

The survey was created to allow cat owners to enter and submit the petting log information, and included additional questions about the cat, household, and owner. First, cat owner participants were asked to confirm that they had completed a petting log for their cat. They were asked to give information regarding their cat's characteristics (e.g., sex, breed, age, medical conditions), and their activities (e.g., outdoor access, enrichment items). Information regarding the household composition, such as how many adults, children, other pets, was also asked. Next, participants entered all of the information from the petting log into the online survey for all three petting sessions. In addition to the petting log, owners were also asked to report on their cat's behaviour and responses to petting during the previous week and over their cat's entire lifetime, and to provide further detail on how they normally respond to the cat showing warning signs, scratching, or biting during petting. The owners were given full written descriptions of the warning signs to

identify (e.g., hissing, skin twitching, biting without puncture) and what would constitute a scratch or bite (i.e., would result in breaking the skin, with puncture). Lastly, owners were asked general demographic information about themselves, including age, gender, province (open to Canadian residents only), and community type. The full survey is available in the Supplementary Materials.

2.3. Data analysis

Statistical analysis was completed using Stata Statistical Software v.15.1 (StataCorp., College Station, TX, USA).

Survey responses were reviewed prior to data analysis for completion and appropriateness for analysis. Responses had to have at least one complete petting session to be included in the analysis and were removed if they indicated that they did not complete the petting log or did not complete the information regarding the petting sessions.

Logistic regression models were used to evaluate the following outcomes focused on instances of aggressive behaviour: 1) in the past week, did the cat scratch or bite the owner while petting; 2) in the time they have owned their cat, has the cat ever scratched or bitten the owner while petting; 3) in the past week, did the cat scratch or bite someone who is not the owner while petting; and 4) in the time they have owned their cat, has the cat ever scratched or bitten someone who is not the owner while petting.

Mixed logistic regression models were used to evaluate cat responses to the petting sessions. Responses to petting sessions were recoded to 1 if the cat's response to petting included the options "Enjoys" or "Accepts", but did not include the other remaining options, and were recoded to 0 if the cat's response to petting included "Resists", "Cannot continue", "Warning behaviours", or "Scratched or bit, breaking the skin", and could also include the previous options. The random effect of "Participant ID" was also included in the mixed logistic regression models to account for variation within participants over the three petting sessions.

Spearman's Rank Correlation and Phi coefficients were used to identify any pairwise correlations between two independent variables. If two variables were deemed highly correlated (rho $\geq$ |0.8|), considerations were made to remove one of the variables based on biological plausibility (Dohoo et al., 2014). All continuous variables were assessed for the assumption of linearity for the logistic regression models. They were assessed graphically using locally weighted scatterplot smoothing curves (lowess curves) against the log odds of each outcome variable. If the relationship was linear, the variable was maintained as a continuous variable. If it appeared to be non-linear, it was modelled as a quadratic relationship if appropriate, otherwise it was categorized based on biologically plausible cut-points or quartiles. The categories generated for the previous continuous variables can be found in the Supplementary Materials.

All variables (Supplementary Materials) were tested individually with each of the four logistic regression models assessing aggressive outcomes to the owner and unfamiliar individuals. For the mixed regression models assessing cat responses to the petting sessions, all independent variables were tested except the owner responses to unwanted behaviours. Variables were retained for further analysis using a liberal significance level (α=0.2) (Dohoo et al., 2014). A manual, forward, stepwise method was used to test the significance of the retained independent variables. Variables were added from highest to lowest levels of significance from univariable analysis and were retained in the model using a significance level of 5% (α =0.05). Variables removed from the model were subsequently examined to determine if they acted as explanatory antecedents or distorter variables (i.e., confounding variables). These variables were considered to have a confounding effect and were retained in the model if they were not intervening variables and their removal resulted in a $\geq$ 30% change to a statistically significant coefficient.

A Hosmer-Lemeshow goodness of fit test for binary data was used to assess the fit of the final logistic regression models (Dohoo et al., 2014; Hosmer and Lemeshow, 1989). Diagnostics were performed to identify any outlying observations, or observations having high leverage or influence on the model. The impacts of these observations on the model was assessed and the observations were inspected for recording errors.

The fit of the mixed logistic regression model was determined by evaluating the assumptions of normality and homoscedasticity of the best linear unbiased predictions (BLUPs) (Dohoo et al., 2014). This was accomplished using normal quantile plots (normality) and scatter plots (homoscedasticity) of the BLUPs against the log odds of the outcome. Outlying observations were detected using Pearson residuals and were evaluated further for any recording errors. The latent variable technique was used to estimate the intra-class correlation coefficient within participants (Dohoo et al., 2014).

3. Results

3.1. Study demographics

A total of 66 cat owner participants completed the petting log survey and were included in the study. A high proportion of cats in this study were female (63.6%), were not declawed (86.4%), were between 7 and 10 years old (28.8%), and were a domestic (90.9%) short-haired (80.3%) cat (Table 1, and Supplementary Materials). All cats in this study were spayed or neutered. Most participants identified as a woman (82.8%), were between 26 and 33 years old (38.2%), and lived in an urban community (63.8%) in Ontario, Canada (74.1%). A summary of cat and participant demographics can be found in Table 1, and full descriptive statistics are available in the Supplementary Materials.

In general, most owners reported that their cat enjoyed the petting session, over all three sessions (82.7–90.6%) (Table 2). Some owners reported that their cat showed warning behaviours over the three petting sessions (10.9–12.7%), and only one cat scratched or bit the owner during the study (Table 2). In petting sessions 1 and 3, owners reported that the cat was more likely to initiate the petting session; however, during petting session 2 it was split between the owner and cat initiating the petting session. On average, owners performed three petting sessions in each time block, and each petting session tended to last approximately 6.5 min (Table 3). The areas cat owners petted the most included: the top of the head, back of the neck, the chin/cheeks, and the upper back (Table 2). Conversely, the least petted areas included: the stomach, chest/throat, and the lower back, near the base of the tail. Full descriptive statistics of each petting session are available in the Supplementary Materials.

3.2. Risk factor analysis

Correlation analysis found three pairs of variables that were highly correlated: 1) are there additional pets in the household, and are there additional cats in the household (rho=0.81); 2) stop petting or grooming as a correction technique in response to warning signs shown during petting, and does not apply for a correction technique used in response to warning signs shown during petting (rho=|0.89|); and 3) is the cat declawed, and providing enrichment access of scratching areas (rho=|0.90|). The decision was made to exclude the following variables for inclusion in multivariable models: 1) are there additional pets in the household; 2) does not apply for a correction technique used in response to warning signs shown during petting; and 3) providing enrichment access of scratching areas.

3.3. History of petting responses

When evaluating the outcomes "in the past week, did the cat scratch or bite the owner while petting" and "in the past week, did the cat scratch or bite someone who is not the owner while petting", no independent variables assessed had a significant association with these outcome variables.

Table 1
Descriptive statistics from the initial survey concerning owner and cat demographics, household characteristics, petting frequencies and cat responses to petting (N = 66).

Variable	Categories	Total (n)	%
Cats owned	1	31	47.0
	2	26	39.4
	3 +	9	13.6
Sex of the cat	Female	42	63.6
	Male	24	36.4
Age of the cat	1–2	7	10.6
	3–6	19	27.3
	7–10	22	28.8
	11 +	18	33.3
Average hours left alone	0	8	12.1
	1–3	17	25.8
	4–7	20	30.3
	8 +	21	31.8
Other cats in the household	No	30	45.5
	Yes	36	54.5
Owner age	18–25	4	7.3
	26–33	21	38.2
	34–41	8	14.5
	42–49	10	18.2
	50–60	8	14.6
	61–70	3	5.5
	Over 70	1	1.8
Gender	Woman	48	82.8
	Man	6	10.3
	My gender identity is not listed	2	3.5
	Prefer not to answer	2	3.5
Province	Ontario	43	74.1
	Outside Ontario	15	25.9
Total petting sessions completed	1	5	7.6
	2	14	21.2
	3	47	71.2
Scratched/bit owner in past week during	No	55	94.8
petting	Yes	3	5.2
Scratched/bit non-owner in past week during	No	58	98.3
petting	Yes	1	1.7
Scratched/bit owner in lifetime during	No	40	67.8
petting	Yes	19	32.2
Scratched/bit non-owner in lifetime during	No	45	76.3
petting	Yes	14	23.7
Responses to warning signs			
Stop petting or grooming	No	15	25.9
	Yes	43	74.1
Verbal correction	No	40	69.0
	Yes	18	31.0
Responses to being scratched or bitten			
Stop petting or grooming	No	36	64.3
	Yes	20	35.7
Verbal correction	No	44	78.6
	Yes	12	21.4

When examining the outcome "in the time you have had your cat, has your cat ever scratched or bitten you (the owner), breaking the skin while petting", the odds of being scratched or bitten in this context were significantly greater when owners reported using a verbal correction (e.g., "No!") in response to being scratched or bitten during petting (Table 4). Additionally, when owners reported that they stopped petting or grooming in response to being scratched or bitten during petting, the odds of being scratched or bitten during petting was significantly greater than those that did not take this action (Table 4). Lastly, owners who reported providing explorative enrichment (boxes, tunnels) had significantly lower odds of being scratched or bitten during petting compared to those that did not provide this enrichment item (Table 4). The Hosmer-Lemeshow goodness of fit test was non-significant indicating that there was no evidence that the model did not fit the data ($p = 0.77$).

Relative to the outcome "in the time you have had your cat, has your cat ever scratched or bitten another individual (not the owner), breaking the skin while petting", the odds of being scratched or bitten in this context were significantly greater when the owner-reported that they stopped petting or grooming in response to being scratched or bitten during petting (compared to those that reported they did not stop) (OR= 8.00; 95% CI: 2.1–31.2: $p = 0.003$). Again, the Hosmer-Lemeshow goodness of fit test was non-significant, indicating that there was no evidence that the model did not fit the data ($p > 0.99$).

3.4. *Current petting responses*

For the mixed logistic regression model with the outcome of response of the cat to the petting sessions in the study, the odds of a positive response ("Enjoys" or "Accepts") were decreased when the owner-reported providing feeding devices (puzzle feeders, food hiding toys) as a form of enrichment (OR= 0.12; 95% CI: 0.02–0.91; $p = 0.041$). The BLUPS had a normal distribution and were homoscedastic. No outliers were identified. The ICC for responses within a participant was 66% (95% CI: 34–88%).

4. Discussion

Most owners in this study reported that their cat responded positively to petting by their owner in the petting sessions. While some owners reported that their cat displayed warning behaviours during these sessions, no significant associations were found between this response and petting body regions, home environment, or owner/cat demographics.

In the current study, owners petted their cats the most on the top of their head, on their chin/cheeks, and on back of their neck, and tended to avoid the areas near the base of the tail, on their chest/throat and on their stomach. The regions owners chose to pet their cat in this study align with regions that have been recommended in previous research; specifically, owners in the current study petted their cats in areas previously described as regions cats enjoy (top of the head, chin/cheeks), and avoided areas previously described as regions cats dislike (base of the tail) (Ellis et al., 2015; Soennichsen and Chamove, 2002). We did not find any associations between petting regions and responses by the cat; however, few cats responded negatively to the petting sessions, so we likely did not have sufficient power to detect an effect. It is possible that owners were less likely to pet the cat in the areas that have previously been associated with negative responses. To fully understand cat responses to petting by body region it would be necessary to have owners pet their cats in each area and record their response. However, for the purposes of this study, we wanted to observe natural owner-cat petting interactions in routine conditions to understand typical cat-owner interactions around petting. Further studies with larger sample sizes are necessary to fully explore this topic.

While a majority of cats responded positively to petting in the current study, it is concerning that a relatively high percentage of owners reported that their cat displayed warning signs (10.9–12.7%) during the petting sessions. Warning signs described in this study were easy to detect and understand, and included hissing, growling, yowling, skin twitching, rapid tail movements, staring, swatting, and biting without puncture. These can be thought of as precursors to potentially harmful aggressive behaviours, including scratching (breaking the skin) and biting (with puncture). In addition, one third of owners reported having been scratched or bitten by their cat in the cat's lifetime, suggesting that in some circumstances these early warning signs might have been missed leading to escalation of the cat's response. In previous studies, when owners were educated on best practice handling during cat-owner interactions (observing cat body language, limiting touch and provision of control), cats showed more affiliative and positive behaviours during their interactions (Haywood et al., 2021). Therefore, if owners were

Table 2
Descriptive statistics for petting during three sessions occurring during the 1st (n = 64), 2nd (n = 52) and 3rd (n = 58) 6-hour period of their waking time of categorial variables.

Categorical		Session 1		Session 2		Session 3	
Variables	Categories	Total (n)	%	Total (n)	%	Total (n)	%
Responses	Enjoys	58	90.6	43	82.7	46	83.6
	Accepts	20	31.3	16	30.8	18	32.7
	Resists	2	3.1	1	1.9	2	3.6
	Cannot continue	4	6.3	1	1.9	4	7.3
	Warning behaviours	7	10.9	6	11.5	7	12.7
	Scratched or bit	1	1.6	0	0.0	0	0.0
Locations	1- top of the head	59	92.2	46	92.0	48	88.9
	2- temporal region	42	65.6	34	66.7	41	75.9
	3- chin/cheeks	49	76.6	35	68.6	38	70.4
	4- back of neck	54	84.4	42	82.4	48	88.9
	5- upper back	47	73.4	39	76.5	42	77.8
	6- middle back	44	68.8	34	66.7	37	68.5
	7- lower back, near base of tail	39	60.9	34	66.7	32	59.3
	8- chest/throat	29	45.3	24	47.1	26	48.2
	9- stomach	12	18.8	19	37.3	12	22.2
Who initiated	Owner	18	28.1	26	51.0	21	38.9
	Cat	46	71.9	25	49.0	33	61.1
Was it a	Yes	64	100.0	48	96.0	54	100.0
normal petting session	No	0	0.0	2	4.0	0	0.0

Table 3
Descriptive statistics of continuous variables for petting during three sessions occurring during the 1st (n = 64), 2nd (n = 52) and 3rd (n = 58) 6-hour period of their waking time.

	Session 1			Session 2			Session 3		
Continuous Variables	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean
How long	0 s	2423 s	397.7 s	0 s	1500 s	286.2 s	0 s	3600 s	493.5 s
Time of day	4:00	19:00	7:52	9:00	19:00	14:43	11:00	23:59	22:38
Total petting sessions	1	21	3.4	1	12	2.8	1	16	3.4
Total hours home	< 1 h	6	5.2	< 1 h	6	4.8	3	6	5.5

educated on best practice petting preferences of cats there could be a reduction in petting responses that include warning behaviours, which could in turn result in a reduction in scratches and/or bites.

Verbal corrections and stopping petting in response to being scratched or bitten during petting were associated with an increased odds of an aggressive response to petting over a cat's lifetime. In a previous study, we found a similar association between aversive correction methods in response to an unwanted behaviour by cats, and aggressive behaviour; however, in both the previous study and the current study the direction of this relationship is unknown (O'Hanley et al., 2021). It is possible that owners are more likely to have the opportunity to show these responses with cats that tend to respond aggressively to petting. Reward-based training to positively reinforce desirable behaviors is recommended to cat owners as it may increase the frequency of desired behaviors, and, consequently, reduce the occurrence of undesirable behaviors (Gazzano et al., 2015; Rodan, 2010). Given that aversive correction techniques can cause increased stress, promote a negative emotional state and may contribute to the escalation of behaviours into an aggressive response, as seen in dogs, it is important to promote reward-based training in our companion animals (Casey et al., 2021; Herron et al., 2009; Hsu and Sun, 2010; Rodan, 2010; Tami et al., 2008). Removing the act of petting during an interaction when the cat demonstrates it is not enjoying the interaction is the best practice response by the owner, and again, training to this effect might help reduce negative responses to petting.

Provision of explorative enrichment (hiding boxes, tunnels) was found to be protective against aggressive responses during petting over a cat's lifetime. In previous literature conducted on cats in a shelter, it was shown that providing cats with hiding spaces was associated with reduction of stress, in a quicker amount of time, compared to those that did not have hiding spaces, as well as a preference for this space over other environmental enrichment items (Ellis et al., 2017; Kry and Casey,

Table 4
Logistic regression model results for the outcome question of: "In the time you have had your cat, has your cat ever scratched or bitten you (the owner), breaking the skin, while petting?" (N = 64).

Variable	Category	OR	p-value	95% CI Lower	95% CI Upper
Using a verbal correction in response to being scratched or bitten during petting	No	Ref.			
	Yes	14.21	0.007	2.16	93.67
Stop petting or grooming in response to being scratched or bitten during petting	No	Ref.			
	Yes	6.90	0.023	1.37	34.76
Having explorative enrichment (boxes, tunnels)	No	Ref.			
	Yes	0.15	0.016	0.03	0.81

2007; Van Der Leij et al., 2019; Vinke et al., 2014). When laboratory cats were exposed to a presumed stressor, it was found that these cats spent more time hiding than those that were not exposed to these stressors (Carlstead et al., 1993). In situations where space is limited (i.e., shelters and laboratory settings), floor space allocation is an important consideration: providing hiding structures may reduce available floor space, and reduced floor space has been linked to fewer affiliative and maintenance behaviors (e.g., approaching, rubbing, eating, grooming; Stella et al., 2014) and higher rates of illness (e.g., feline upper respiratory disease; Wagner et al., 2018). Providing cats with a space to feel safe and protected may help the cat avoid or escape a potentially stressful interaction or feel less threatened, leading to them being less likely to display aggressive responses. Therefore, providing explorative enrichment, including spaces to hide (while considering spacing restrictions), can potentially be used as an easy enrichment option to provide cats with a safe space that might help reduce stress and aggressive occurrences. Further controlled studies are needed to explore the effects of this enrichment on cat stress and aggression.

The only factor that was found to be significantly associated with the cat having an overall "positive" or "negative" response to petting during the study sessions was when owners reported providing food puzzle enrichment for their cat. However, the response was opposite to what we had predicted, with these cats being less likely to have an overall "good" response to petting during the petting sessions. The impact of food puzzles on cat welfare has not been fully evaluated, although they are commonly encouraged as a form of enrichment and assumed to promote positive welfare. Two studies have explored this relatively new form of enrichment for cats and have evaluated the practices and use of food puzzles, as well as the preferences in how the food is offered (freely available vs. requiring effort) (Delgado et al., 2020; 2021). Researchers found that cats preferred freely available food over having to work for their food presented in a food puzzle (Delgado et al., 2021), and this is unlike other species which tend to prefer working for their food over free feeding (Jensen, 1963; Lindqvist et al., 2002). The preference for free feeding over food puzzles may contribute to frustration when food puzzles are offered and impact a cat's responses to this owner-cat interaction. However, to test this hypothesis, further research is needed to fully understand the efficacy of food puzzles as a form of positive enrichment for cats and potential impacts on cat-owner interactions and aggression.

4.1. Limitations

The sample size for the current study was relatively small and the cats included in the sample showed fewer negative responses during petting sessions than what we had originally anticipated. As reported in this study, most owners stated that their cat enjoyed petting during the petting sessions (82.7%-90.6%). When this study was distributed, potential participants were informed that they would need to complete a petting log describing petting sessions with their cat. Because we did not target our advertisements to encourage owners who had cats that did not enjoy petting to complete this study, we may have obtained a biased sample of cats that enjoy petting. Owners that did have cats that do not generally respond positively to petting may have been less inclined to participate because they are unable to pet their cat without a negative response.

This study was released in July 2020, during the COVID-19 pandemic in an online format. The pandemic resulted in an increase in the number of people working from home, so it is likely that the owners' lifestyles may have been altered, and this might have affected cat behaviour.

This study was limited to participants residing in Canada and the United States, and our study mainly included participants from Ontario, Canada, who were under 50 years old and identified as female. Most cat owners in Canada are under the age of 55 so this is unlikely to be a concern. However, previous literature has shown cross-cultural differences in human-cat interactions, including attitudes towards pet cats (Vitale et al., 2024), thus the results of these studies may not be reflective of human-cat interactions outside of North America. Additionally, in North America, the gender of cat owners is fairly evenly distributed between men and women, so our study underrepresented male cat owners (Humane Canada, 2017; Sax et al., 2003; Statista, 2016). This is a notable limitation because inherent gender differences have been found in cat-human relationships (Demirbaş et al., 2026), meaning our results may have limited generalizability to male cat owners.

Lastly, due to the nature of cross-sectional studies, any results obtained cannot infer causation and we can only report on associations between factors. The results presented in this study can be used to help inform future studies surrounding owner-cat interactions, and petting preferences and responses of domestic cats.

5. Conclusion

While most owners participating in the current study reported that their cat enjoys petting, we still found that many cats displayed warning signs during petting sessions, and a high percentage of owners reported being bitten or scratched during petting, in the cat's lifetime. Although we did not find associations between cat responses and cat petting regions, we did find associations between petting responses and enrichment options and training methods. This suggests that good cat welfare in the home is linked to improved cat-human interactions and reduced aggression during petting, and these areas require further assessment in future research. These findings emphasize the need for the use of educational training tools to help owners better understand cat needs and preferences, with an emphasis on appropriately responding to cat warning behaviours and provision of environmental enrichment.

CRediT authorship contribution statement

Kristina O'Hanley: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David L. Pearl:** Writing – review & editing, Visualization, Supervision, Software, Methodology, Formal analysis. **Mary Klinck:** Writing – review & editing, Visualization, Validation, Supervision, Methodology. **Lee Niel:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Funding

Funding for this project was obtained by Dr. Lee Niel from the Natural Sciences and Engineering Research Council of Canada (NSERC; grant number RGPIN-2019-06012). Stipend support for Dr. Kristina O'Hanley was provided by the University of Guelph Ontario Veterinary College Graduate Stipend, the Queen Elizabeth II Graduate Scholarship in Science and Technology, and the Ontario Graduate Scholarship.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank all participants for their time and contribution to this study.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.applanim.2026.107209.

References

Behnke, A.C., Vitale, K.R., Udell, M.A.R., 2021. The effect of owner presence and scent on stress resilience in cats. Appl. Anim. Behav. Sci. 243. https://doi.org/10.1016/j.applanim.2021.105444.

Bernstein, P.L., 2000. People petting cats: a complex interaction. Abstr. Anim. Behav. Soc. Annu. Conf. 9.

Bernstein, P.L., 2007. The human-cat relationship. In: Rochlitz, I. (Ed.), The Welfare of Cats, 3rd ed. Springer Press, pp. 44–89. https://doi.org/10.1201/b21911.

Bradshaw, J.W.S., Casey, R.A., Brown, S.L., 2012. The Behaviour of the Domestic Cat, 2nd ed.). CABI.

Carlstead, K., Brown, J.L., Strawn, W., 1993. Behavioral and physiological correlates of stress in laboratory cats. Appl. Anim. Behav. Sci. 38 (2), 143–158. https://doi.org/10.1016/0168-1591(93)90062-T.

Casey, R.A., Naj-Oleari, M., Campbell, S., Mendl, M., Blackwell, E.J., 2021. Dogs are more pessimistic if their owners use two or more aversive training methods. Sci. Rep. 11 (1), 1–9. https://doi.org/10.1038/s41598-021-97743-0.

Delgado, M., Bain, M.J., Buffington, C.T., 2020. A survey of feeding practices and use of food puzzles in owners of domestic cats. J. Feline Med. Surg. 22 (2), 193–198. https://doi.org/10.1177/1098612X19838080.

Delgado, M.M., Han, B.S.G., Bain, M.J., 2021. Domestic cats (Felis catus) prefer freely available food over food that requires effort. Anim. Cogn. 0123456789. https://doi.org/10.1007/s10071-021-01530-3.

Demirbaş, Y.S., Kerman, K., Atılgan, D., Ünler, M., Yildirim, T., Şimşek, S., 2026. Greeting Vocalizations in Domestic Cats Are More Frequent With Male Caregivers. Ethology 132 (2), 87–94.

Dire, D.J., 1991. Cat bite wounds: Risk factors for infection. Ann. Emerg. Med. 20 (9), 973–979. https://doi.org/10.1016/S0196-0644(05)82975-0.

Dohoo, I., Martin, W., Stryhn, H., 2014. Veterinary Epidemiologic Research, 2nd ed.). VER Inc.

Ellis, J.J., Stryhn, H., Spears, J., Cockram, M.S., 2017. Environmental enrichment choices of shelter cats. Behav. Process. 141 (October 2016), 291–296. https://doi.org/10.1016/j.beproc.2017.03.023.

Ellis, S.L.H., Thompson, H., Guijarro, C., Zulch, H.E., 2015. The influence of body region, handler familiarity and order of region handled on the domestic cat's response to being stroked. Appl. Anim. Behav. Sci. 173, 60–67. https://doi.org/10.1016/j.applanim.2014.11.002.

Fritz, J.N., Fletcher, V.L., Dyer, S.P., Carpenter, K.A., Skrbec, M.L., Yankelevitz, R.L., 2022. Functional analysis and treatment of aggression exhibited by cats toward humans during petting. J. Appl. Behav. Anal. 55 (1), 169–179. https://doi.org/10.1002/jaba.877.

Gazzano, A., Bianchi, L., Campa, S., Mariti, C., 2015. The prevention of undesirable behaviors in cats: Effectiveness of veterinary behaviorists' advice given to kitten owners. J. Vet. Behav. Clin. Appl. Res. 10 (6), 535–542. https://doi.org/10.1016/j.jveb.2015.07.042.

Gourkow, N., Hamon, S.C., Phillips, C.J.C., 2014. Effect of gentle stroking and vocalization on behaviour, mucosal immunity and upper respiratory disease in anxious shelter cats. Prev. Vet. Med. 117 (1), 266–275. https://doi.org/10.1016/j.prevetmed.2014.06.005.

Haywood, C., Ripari, L., Puzzo, J., Foreman-Worsley, R., Finka, L.R., 2021. Providing Humans With Practical, Best Practice Handling Guidelines During Human-Cat Interactions Increases Cats' Affiliative Behaviour and Reduces Aggression and Signs of Conflict. Front. Vet. Sci. 8 (July). https://doi.org/10.3389/fvets.2021.714143.

Herron, M.E., Shofer, F.S., Reisner, I.R., 2009. Survey of the use and outcome of confrontational and non-confrontational training methods in client-owned dogs showing undesired behaviors. Appl. Anim. Behav. Sci. 117 (1–2), 47–54. https://doi.org/10.1016/j.applanim.2008.12.011.

Hosmer, D., Lemeshow, S., 1989. Applied Logistic Regression. Wiley.

Hsu, Y., Sun, L., 2010. Factors associated with aggressive responses in pet dogs. Appl. Anim. Behav. Sci. 123 (3–4), 108–123. https://doi.org/10.1016/j.applanim.2010.01.013.

Humane Canada. (2017). Cats in Canada: A five-year review of cat overpopulation.

Jensen, G.D., 1963. Preference for bar pressing over "freeloading" as a function of number of rewarded presses. J. Exp. Psychol. 65 (5), 451–454. https://doi.org/10.1037/h0049174.

Kry, K., Casey, R., 2007. The effect of hiding enrichment on stress levels and behaviour of domestic cats (Felis sylvestris catus) in a shelter setting and the implications for adoption potential. Anim. Welf. 16 (3), 375–383.

Lindqvist, C.E.S., Schutz, K.E., Jensen, P., 2002. Red Jungle Fowl Have More Contrafreeloading than White Leghorn Layers: Effect of Food Deprivation and Consequences for Information Gain. Behaviour 139 (9), 1195–1209.

O'Hanley, K.A., Pearl, D.L., Niel, L., 2021. Risk factors for aggression in adult cats that were fostered through a shelter program as kittens. Appl. Anim. Behav. Sci. 236, 1–10. https://doi.org/10.1016/j.applanim.2021.105251.

Palacio, J., León-Artozqui, M., Pastor-Villalba, E., Carrera-Martín, F., García-Belenguer, S., 2007. Incidence of and risk factors for cat bites: a first step in prevention and treatment of feline aggression. J. Feline Med. Surg. 9 (3), 188–195. https://doi.org/10.1016/j.jfms.2006.11.001.

Ramos, D., Mills, D.S., 2009. Human directed aggression in Brazilian domestic cats: owner reported prevalence, contexts and risk factors. J. Feline Med. Surg. 11 (10), 835–841. https://doi.org/10.1016/j.jfms.2009.04.006.

Rodan, I., 2010. Understanding Feline Behavior and Application for Appropriate Handling and Management. Top. Companion Anim. Med. 25 (4), 178–188. https://doi.org/10.1053/j.tcam.2010.09.001.

Sax, L.J., Gilmartin, S.K., Bryant, A.N., 2003. Assessing Response Rates and Nonresponse Bias in Web and Paper Surveys. Res. High. Educ. 44 (4), 409–432.

Soennichsen, S., Chamove, A.S., 2002. Responses of cats to petting by humans. Anthrozoos 15 (3), 258–265. https://doi.org/10.2752/089279302786992577.

Statista, 2016. Share of consumers who own a cat in Canada as of May 2016, by gender. Statista Research Department. https://www.statista.com/statistics/468163/share-of-adults-who-own-a-cat-by-gender-canada/.

Stella, J., Croney, C., Buffington, T., 2014. Environmental factors that affect the behavior and welfare of domestic cats (Felis silvestris catus) housed in cages. Appl. Anim. Behav. Sci. 160 (1), 94–105. https://doi.org/10.1016/j.applanim.2014.08.006.

Tami, G., Barone, A., Diverio, S., 2008. Relationship between management factors and dog behavior in a sample of Argentine Dogos in Italy. J. Vet. Behav. Clin. Appl. Res. 3 (2), 59–73. https://doi.org/10.1016/j.jveb.2007.09.002.

Van Der Leij, W.J.R., Selman, L.D.A.M., Vernooij, J.C.M., Vinke, C.M., 2019. The effect of a hiding box on stress levels and body weight in Dutch shelter cats; A randomized controlled trial. PLoS. ONE 14 (10), 1–14. https://doi.org/10.1371/journal.pone.0223492.

Vinke, C.M., Godijn, L.M., van der Leij, W.J.R., 2014. Will a hiding box provide stress reduction for shelter cats? Appl. Anim. Behav. Sci. 160 (1), 86–93. https://doi.org/10.1016/j.applanim.2014.09.002.

Vitale, K.R., Takagi, S., Arahori, M., Chijiiwa, H., Udell, M.A.R., Kuroshima, H., 2024. A Cross-Cultural Comparison of Cat–Human Relationships in the United States and Japan. AnthrozoöS. 37 (5), 813–831. https://doi.org/10.1080/08927936.2024.2395127.

Wagner, D.C., Kass, P.H., Hurley, K.F., 2018. Cage size, movement in and out of housing during daily care, and other environmental and population health risk factors for feline upper respiratory disease in nine North American animal shelters. PLOS ONE 13 (1), e0190140. https://doi.org/10.1371/journal.pone.0190140.